Luxshan Thuraisingam

Software Engineer

T-Luxshan | Luxshan Thuraisingam | luxshan.thuraisingam@gmail.com | +94 76 454 1834
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Summary

Software Engineer with R&D experience at PickMe. Proven expertise in designing complex systems including a three-layer image forensics system and benchmarked distributed message brokers for peak-load passenger handling. Built ML-driven auto-scaling framework for VMs in cloud reducing energy consumption while meeting SLOs. Expertise in Go, Spring Boot, Docker, Kubernetes, and event-driven systems.

Experience

- **Software Engineer Intern (R&D)** Mar 2025 – Sep 2025
  Digital Mobility Solutions Lanka PLC (**PickMe**), Colombo, Sri Lanka
  - Designed and implemented a three-layer image forensics system to detect tampered driver documents using Error Level Analysis (ELA), JPEG structure decoding, and EXIF metadata analysis; integrated with React frontend, Golang backend, and containerized with Docker.
  - Evaluated and benchmarked Golang-based distributed queue technologies to optimize passenger request handling during peak demand.
  - Developed RESTful API endpoints for QR payment workflow and built React Native components for operational demo application.

Technical Skills

- **Languages & Frameworks:** Go, Java (Spring Boot), Python, React.js, React Native.
- **Backend & Distributed Systems:** Kafka, Redis, RabbitMQ, ETCD, MySQL, RESTful APIs
- **DevOps & Infrastructure:** Docker, Kubernetes GKE, GitHub Actions (CI/CD)
- **Testing & Analysis:** Serenity BDD, Kibana, SonarQube
- **ML & Data:** PyTorch, Scikit-learn, Pandas, Dask, SciPy, Jupyter Notebooks

Education

- **B.Sc. (Hons) in Information Technology** (2022 – 2026)
  University of Moratuwa, Colombo, Sri Lanka
  CGPA - **3.45/4.0**
- **G.C.E. Advanced Level** (2019)
  J/Mahajana College, Tellippalai, Sri Lanka
  Physical Science Stream, Results: 2As 1C, Z-Score: **2.0104**

Research

- **A VM-Aware Energy-Optimal Model Predictive Control Framework for Proactive Horizontal Autoscaling of Virtual Machines** (Ongoing)
  Final Year Research Project (Individual)
  Technologies: Python, PyTorch, Scikit-learn, Pandas, Dask, SciPy, Jupyter Notebooks
  - Developed a hybrid ML and Model Predictive Control (MPC) framework for energy-aware auto-scaling of Virtual Machines in Kubernetes, targeting reduced energy consumption while maintaining acceptable service latency.
  - Built an ML surrogate model pipeline to forecast cloud energy consumption and latency based on underlying VM workload.
  - Implemented a custom MPC receding-horizon controller in Python that uses trained models to optimize VM replica counts over a planning horizon, respecting latency SLOs and stability constraints.

Projects

1. **Message Broker Benchmarking System** - Personal R&D Project (Ongoing)
   Technologies: Golang, Apache Kafka, Redis, RabbitMQ, Docker GitHub
   - Developing a Go-based benchmarking framework to evaluate distributed message brokers under high-concurrency workloads.
   - Implementing multi-producer/multi-consumer simulations to measure throughput, latency, and reliability across Kafka, RabbitMQ, and Redis-based queues.
   - Designing configurable load scenarios and automated experiments to compare performance trade-offs and scalability characteristics.

2. **Distributed Queue Evaluation** - Personal R&D Project
   Technologies: Golang, Redis, RabbitMQ, Docker GitHub
   - Implemented and benchmarked multiple distributed queue technologies to evaluate and compare their performance characteristics.
   - Simulated high-concurrency scenarios to analyze throughput and latency metrics during peak demand cycles.

3. **LabourLINK (Labour Hiring System)** - 2nd Year Group Project
   Technologies: React.js, React Native, Spring Boot, MySQL GitHub
   Client: AlphaCodes (Pvt) Ltd
   - Developed map-integrated on-demand labor hiring web and mobile apps with real-time communication.
   - Contributions (Team Leader & Full-Stack Developer):
     - Implemented authentication & session management using Spring Security and JWT, OTP-based password reset with JavaMailSender.
     - Built profile, bookings, review, rating, and reporting functionalities.
     - Created performance reports with MUI charts and filtering.

4. **LinkNestSocialMedia** - Individual Project
   Technologies: React.js, Spring Boot, MySQL, Docker, GitHub Actions GitHub
   - Developed a full-stack mini social media application supporting posts, likes, and profile management.
   - Implemented JWT-based authentication and secure password reset using JavaMailSender.
   - Set up CI/CD pipelines with GitHub Actions, including automated JUnit testing and Dockerized.

Achievements

- President Awarded Scout (2018)
- Hackathon - XTREME 2023
- Code Rush 2023 - INTECS UoM (15th/120)
- MoraXtreme 8.0 - IEEESB UoM

Extra-Curricular Activities

- **AIESEC in Sri Lanka** (2022 – 2024)
  - Organizing Committee Member - External Relations, Lead CS 9.0 (Nov 2022 – Feb 2023)
  - OC Vice President - External Relations, Beauty Beyond Ceylon (May 2022 – Aug 2022)
- Scout Organization - J/Mahajana College (2009–2019)
- Chess - J/Mahajana College (2013–2019)

Referees

Lahiru Madushanka
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